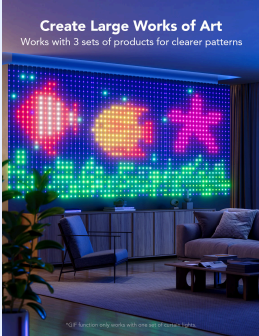
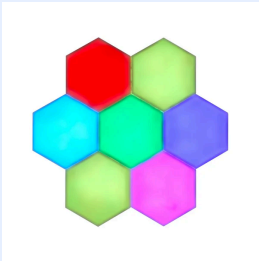


Element B

Existing Lighting Solutions

Name	Image	Description	Strengths	Weaknesses	Relevance
LED curtains		Surrounding sensory experience that allows for a change in environments; large enough for children to stand within	- Very immersive - Unique design that kids would be enamored by	- Lights may affect children with sensitive eyes - There may not be enough space in the special needs - Curtains require being hung up from a specific height, proportionality may be an issue	Children will find this enticing experience calming as it is proven that special needs kids have a need for lighting in order to calm them in sensory-like spaces
LED Light Up Pop Tubes Fidget Toy		It is a sensory toy and fidget tool made with flexible and colorful plastic. It is meant to stretch, bend, and twist. The lights can be put inside to allow for a more fun experience while playing with the toy.	- Interactive - Fun to look at and play with - Gives the kids something to do	- Lights could easily break depending on how aggressively the students play with it - Students could be sensitive to the bright lights	Students in this class are very active and this can be a relaxing toy for them to play with, while also being colorful and interesting to look at.
LUMAGIC LIGHTS		This a cause and effect game that allows kids to explore the sense of touch when exploring differences in colors	-Interactive -Entertaining - Teaches cause and effect - Kids can get up and do something physically with the lights	-Safety issue of security of the fastening of the LED onto the wall - More structured LED lights that cannot really be adjusted to fit different needs	This tool incorporates not only the fun and exciting experience of lighting, but brings in a component of cause and effect.

				within the class	This is something that was specifically requested by teachers and that we know would benefit the kids
LED Cloud Lights		These are LED lights that are shaped in the form of a cloud and hang from the ceiling in order to create an engaging environment and set a mood	- Leans into the aspect of an environment - Fun for kids because they're clouds - Entertains the kids	- Could pose as a hazard since it is dangling down from the ceiling - May not be durable enough to withstand kids constantly messing with or playing with it	This design truly introduces an immersive experience, as it is modeled after clouds and contributes to the idea of a fun and controlled environment for the kids. It is unique and sparks many ideas for unique ways to organize LED lights.
Normal LED Light Strips		Lights that come in different size strips and packs in order to put around your room on the ceiling or wall. They come with a remote and plug into the wall for power; they often connect to phone apps or use bluetooth.	- Customizable for any mood or need - Easy to use - Connectivity to devices/ remote capabilities - Come with timers and lighting patterns	- Limited to what comes with the LED lights– can't be used in any other capacity - Very basic design, not too many features	We will most likely be coding normal LED light strips to create different moods, patterns, and games.

Analysis & Takeaways from Prior Designs

All of the existing designs that were mentioned are unique and beneficial in their own ways. According to the Indiana University of Pennsylvania, LED lights present many benefits overall, such as energy efficiency, durability over time, and flexibility of design. More specifically, lighting solutions like the LUMAGIC lights have cause and effect built into them, while others like the LED Light Up Pop Tubes have a game or toy aspect to them that the kids would enjoy. Other designs like the Curtain Lights or the LED Cloud Lights are simply unique and create an immersive experience for kids that sets them apart from other designs. However, the Massachusetts Institution of Technology highlights that creating your own LED lighting system can make the lights more versatile, more customizable in terms of brightness, and easier to use. The ideal solution and best way to use all of the prior designs for our project is to create a combination of all the aforementioned benefits and put them into one design. This one design would encompass everything that our group loves, and would not include anything that may cause harm or danger towards the students in the room. Our group loves the idea of a full class-surrounding sensory activity to be made with games and toys built into it, so we will be using a little bit of each design in order to create that perfect product for the class.

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